

JANET LIU

janet.liu@queensu.ca | +1 604-363-6812 | linkedin.com/in/liu-janet | <https://janettliu.github.io/>

EDUCATION

Queen's University

Kingston, ON

B.A.Sc in Mechatronics and Robotics Engineering

2024 - 2028

- Dean's Scholar
- Ontario Engineering Competition - Jr. Design, First Place, Canadian Engineering Competition - Jr. Design, Third Place

Canadian Flight Center

Kamloops, BC

Private Pilot's License

2023

- Gained a comprehensive understanding of general aviation and aircraft systems through Air Cadets Power Pilot Scholarship

PROFESSIONAL EXPERIENCE

Advanced Aircraft Design Lab - Royal Military College of Canada

Kingston, ON

Undergraduate Research Assistant

May 2026 - Present

- Led in-flight test campaign of a flow separation mechanism for a 3D-printed UAV, developing the risk assessment, mitigation plan, and analyzing ArduPilot flight data in MATLAB to quantify turn performance gains
- Designed wings for 3D-printed UAVs using MATLAB panel method analysis, iterating on planform, and airfoil selection to meet performance targets
- Prepared a rotorcraft flow simulation model in FlightStream, applying meshing, boundary conditions, and using custom Python scripts to define propeller blade tip path motion

EXTRA-CURRICULAR AND PROJECT EXPERIENCE

Queen's Aerospace Design Team

Aerodynamics Director

Jun 2026 - Present

- Directing the design of a VTOL UAV for SAE Aero 2027, ensuring design rigor, adherence to engineering standards, smooth project milestones and cross-team communication
- Organizing team timeline, including Gantt charts, email workflows, and knowledge-retention archives

Queen's Aerospace Design Team

Aerodynamics Manager

Sep 2025 - May 2026

- Led an 11-member sub-team in designing a fixed-wing UAV for the AIAA 2026 competition, coordinating with structures and systems teams to resolve technical conflicts
- Optimized aircraft aerodynamics using OpenVSP, XFLR5/flow5, custom Python optimization scripts, as well as OpenFOAM CFD to verify stability and performance
- Trained members in CAD, aerodynamic principles, prototyping, and established documentation for design processes
- Resolved design and manufacturing issues during prototyping; aircraft successfully met all mission objectives

Queen's Aerospace Design Team

Research Member

Oct 2024 - Sep 2025

- Executed full OpenFOAM CFD workflow for a NACA0012 airfoil, from geometry preparation, solver setup, turbulence modelling, to post-processing in ParaView
- Validated results to within 12% of experimental data through mesh refinement and boundary condition iteration
- Selected as an AEAC Student Paper Competition finalist and presented at the 16th IMAV international conference

SKILLS

- **Programming & Analysis:** MATLAB, Python, C/C++, Arduino
- **CAD & Design:** SolidWorks, Fusion 360, OpenVSP, 3D printing
- **Aerodynamics & Simulation:** OpenFOAM, Simcenter FlightStream, XFLR5/flow5, Tecplot, ParaView, VLM/panel methods
- **Flight Systems & Testing:** ArduPilot, flight data analysis, flight test planning, risk assessment
- **Tools:** LaTeX, Excel, MS Project, PowerPoint, Word, Notion
- **Certifications:** Restricted Radio Operator Certificate - Aeronautical, Basic Drone License, Standard First Aid & CPR-C
- **Languages:** English (fluent), French (proficient), Mandarin (proficient)